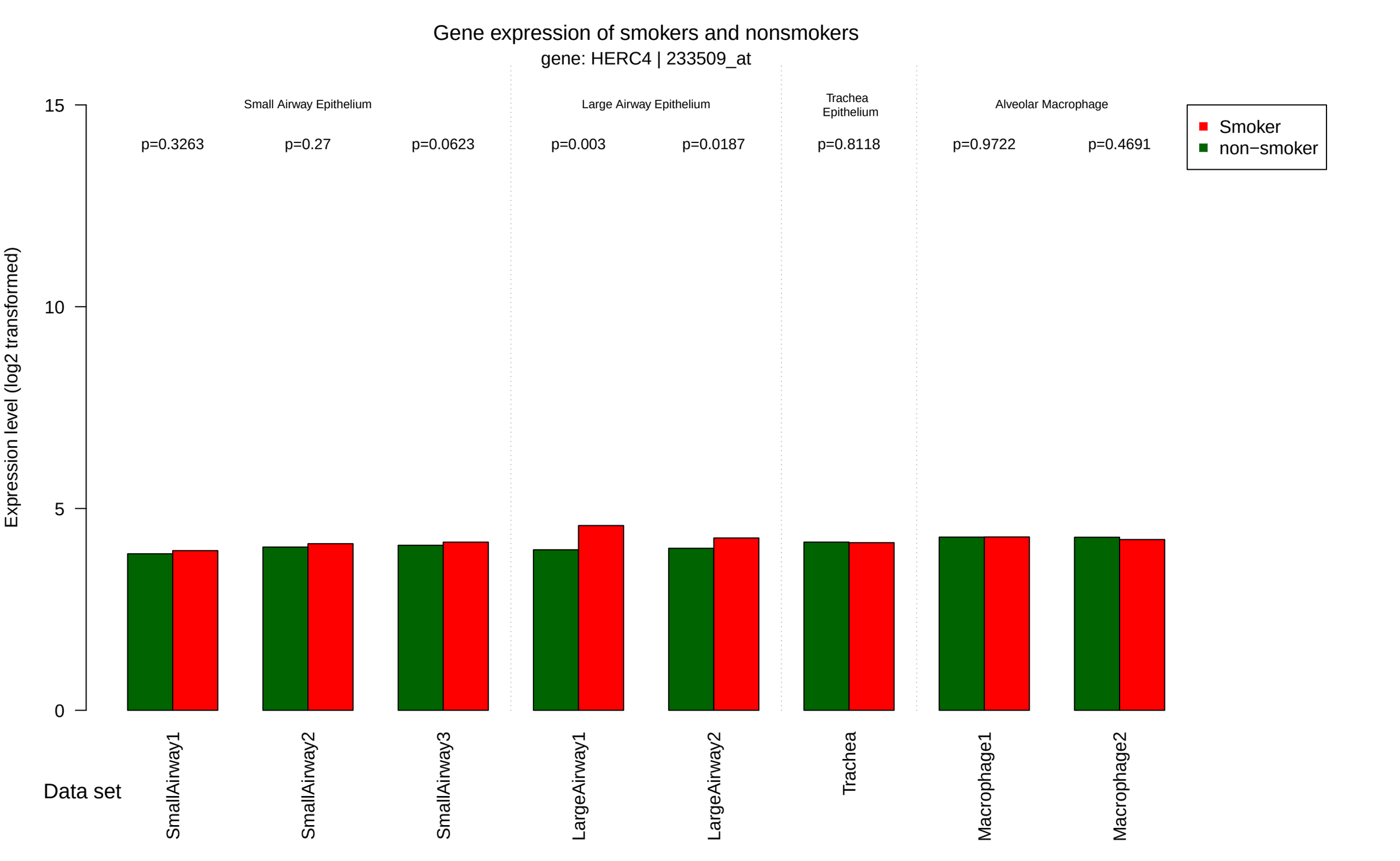
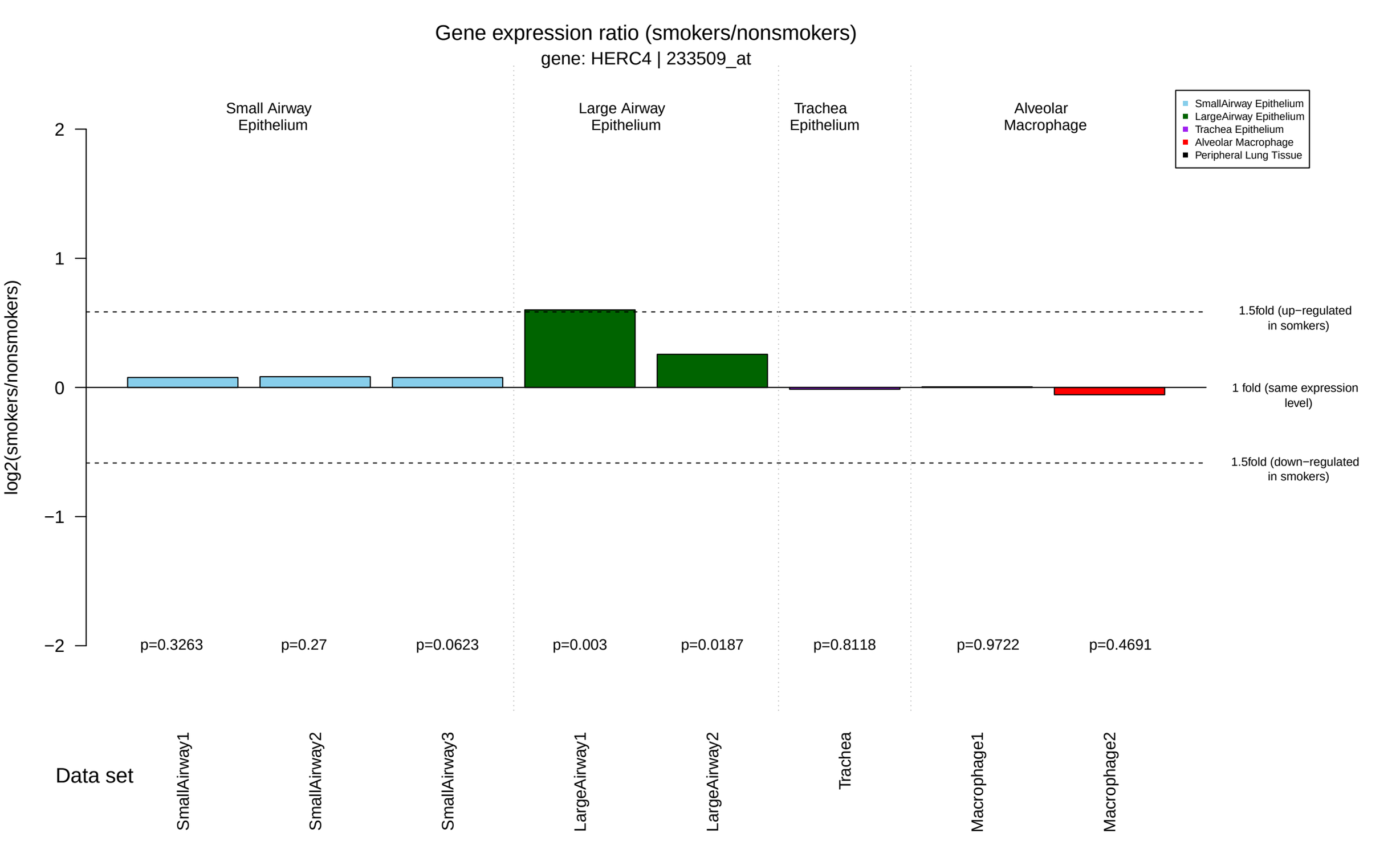
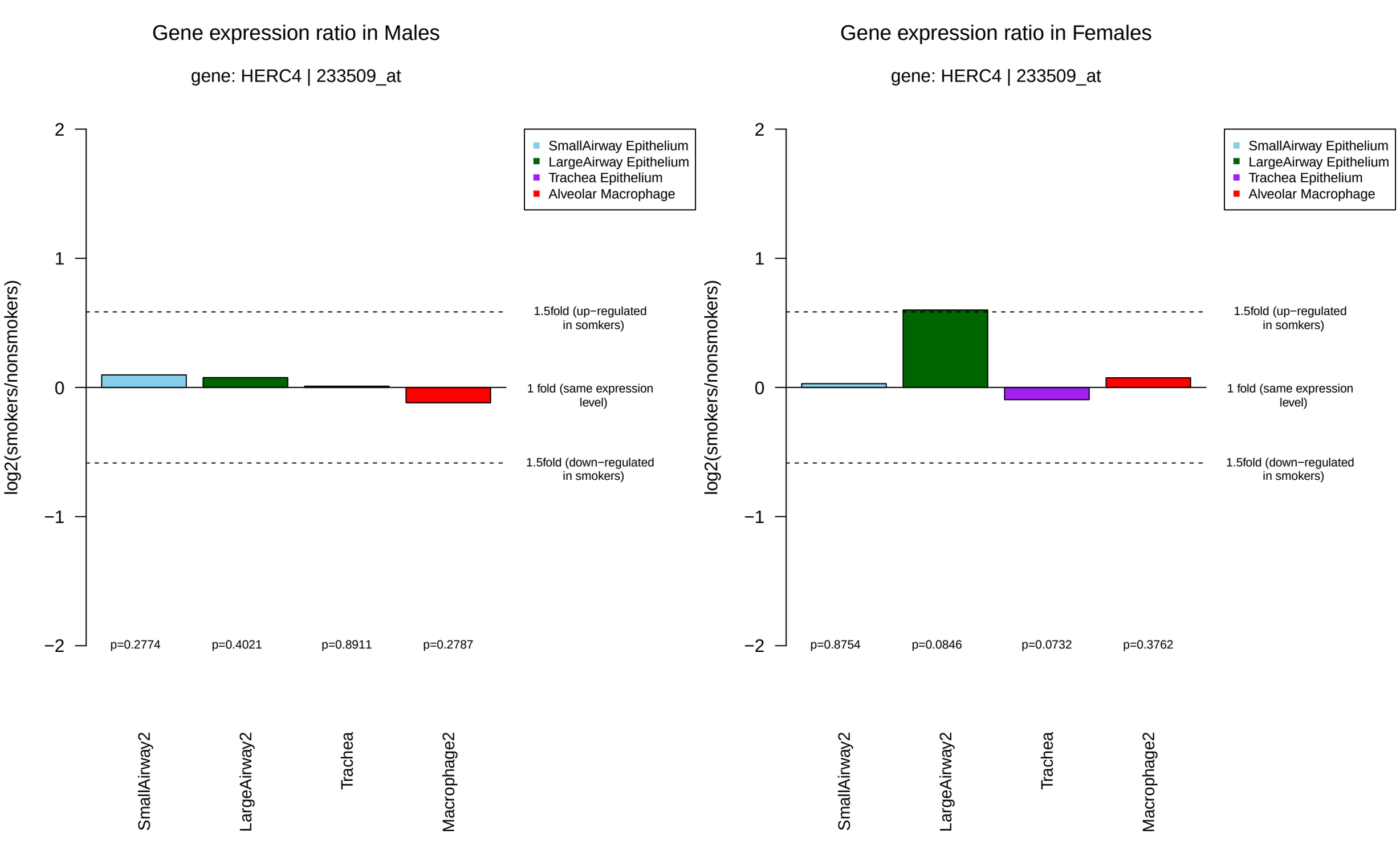
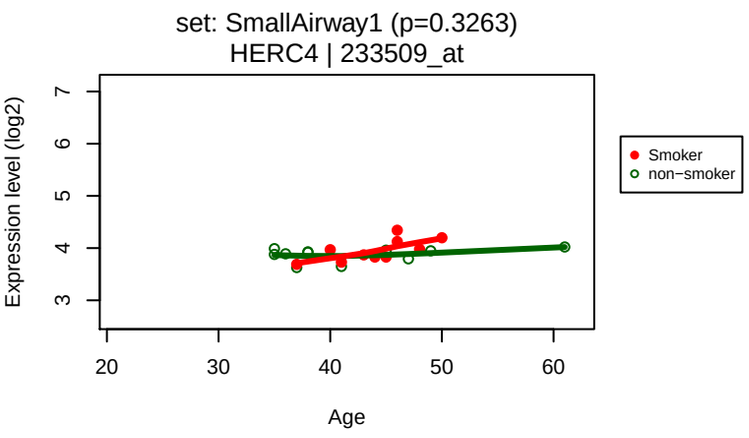
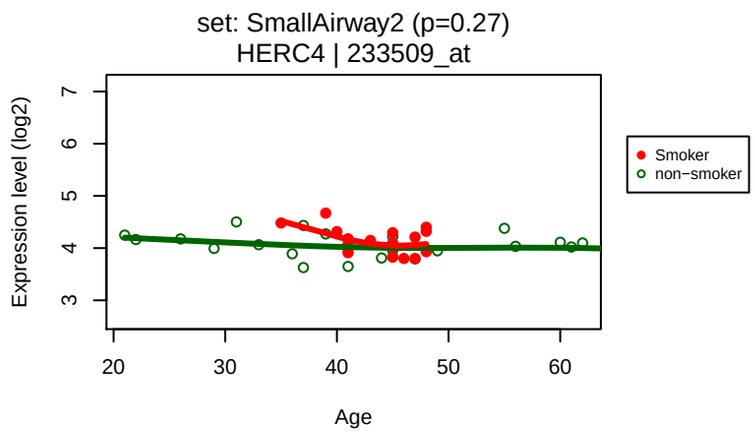
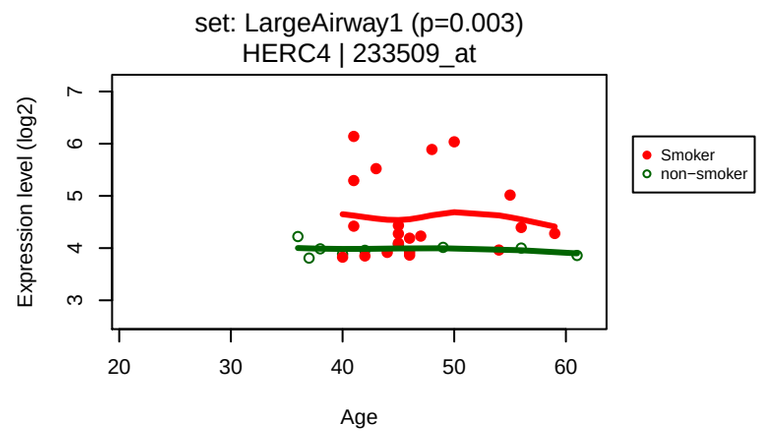

Gene expression of smokers and nonsmokers
gene: HERC4 | 233509_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3263
p=0.27
p=0.0623
p=0.003
p=0.0187
p=0.8118
p=0.9722
p=0.4691
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

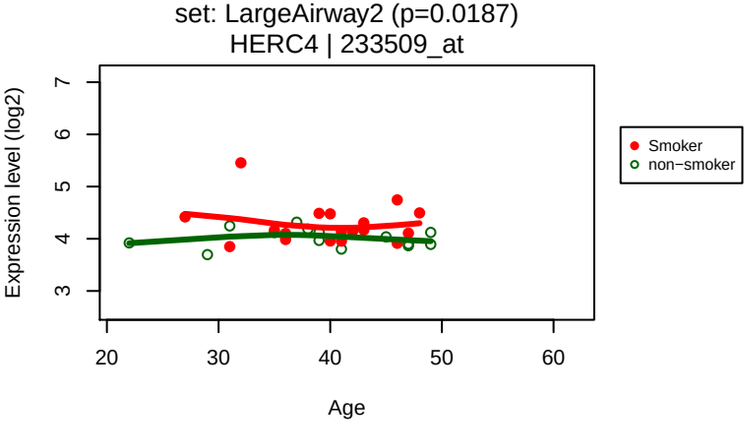

Gene expression ratio (smokers/nonsmokers)
gene: HERC4 | 233509_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3263
p=0.27
p=0.0623
p=0.003
p=0.0187
p=0.8118
p=0.9722
p=0.4691
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

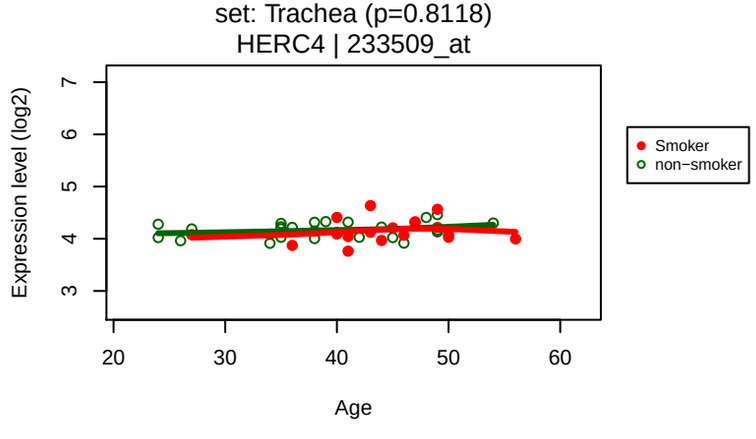

Gene expression ratio in Males
gene: HERC4 | 233509_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2774
p=0.4021
p=0.8911
p=0.2787
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HERC4 | 233509_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8754
p=0.0846
p=0.0732
p=0.3762
SmallAirway2
LargeAirway2
Trachea
Macrophage2

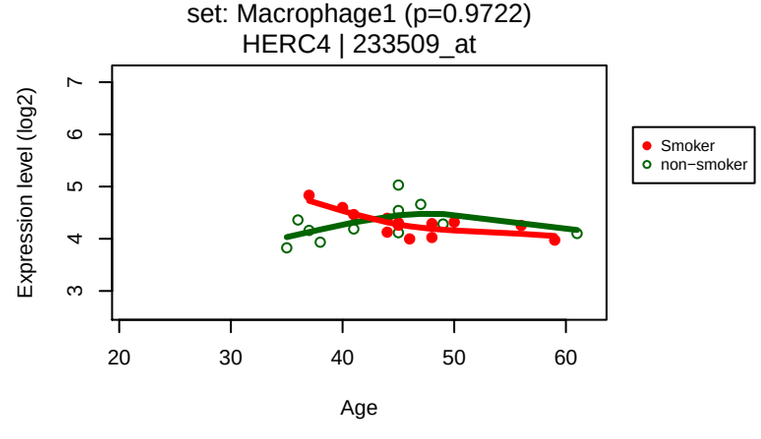

set: SmallAirway1 (p=0.3263)
HERC4 | 233509_at
Expression level (log2)
Age
Smoker
non−smoker

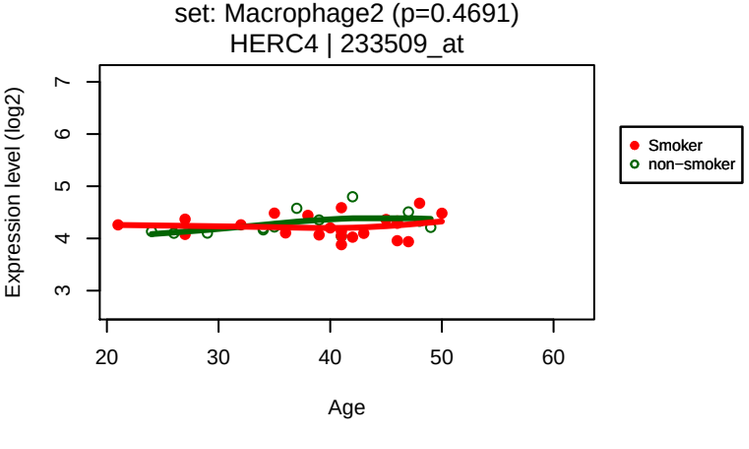

set: SmallAirway2 (p=0.27)
HERC4 | 233509_at
Expression level (log2)
Age
Smoker
non−smoker

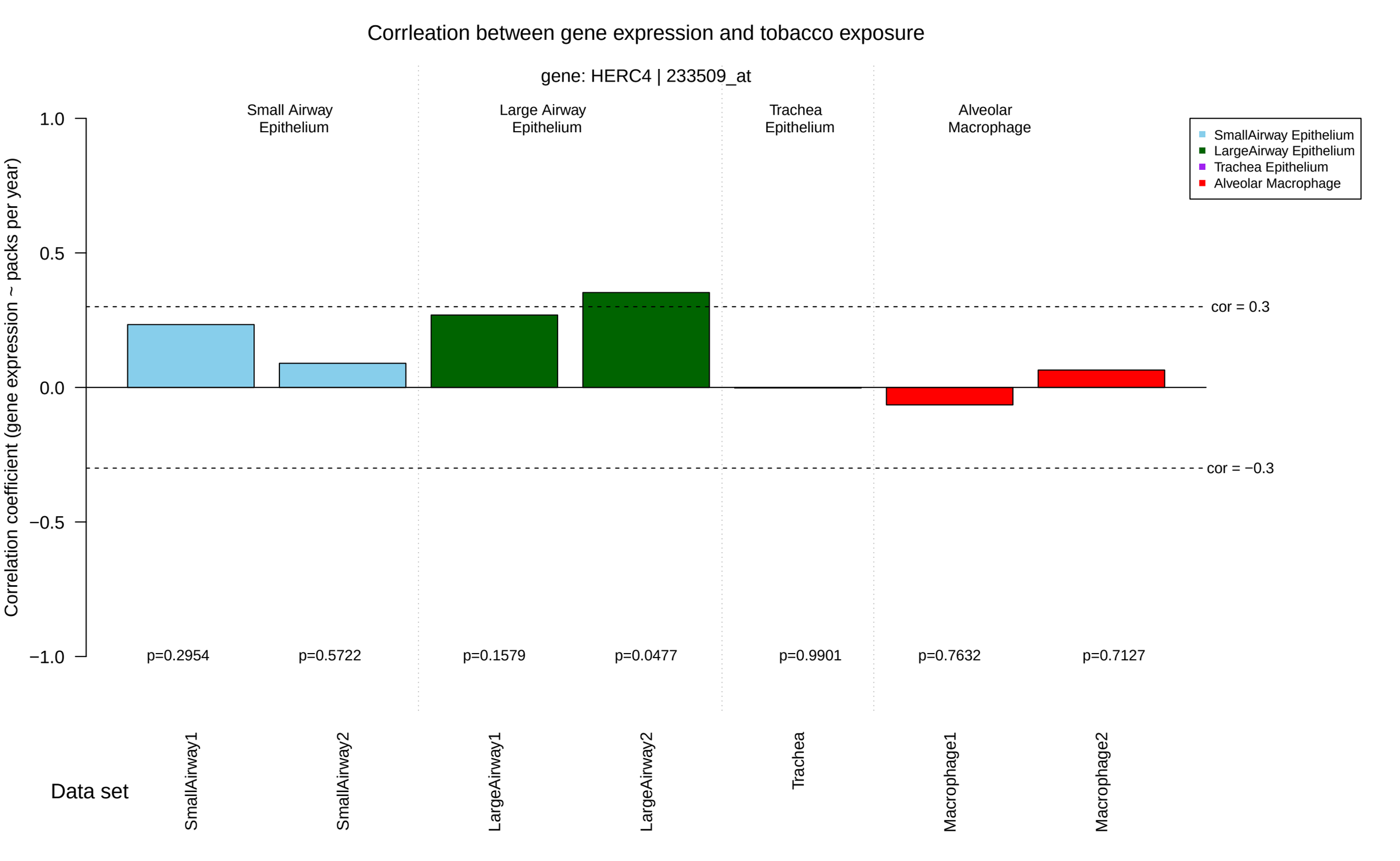

set: LargeAirway1 (p=0.003)
HERC4 | 233509_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0187)
HERC4 | 233509_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8118)
HERC4 | 233509_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9722)
HERC4 | 233509_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4691)
HERC4 | 233509_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HERC4 | 233509_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.2954
p=0.5722
p=0.1579
p=0.0477
p=0.9901
p=0.7632
p=0.7127
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2